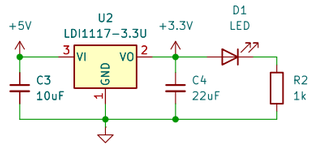
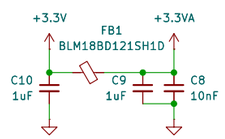


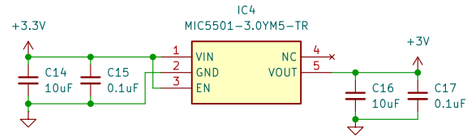
Main 5V regulator



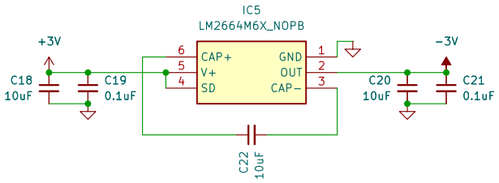
Main 3.3V regulator



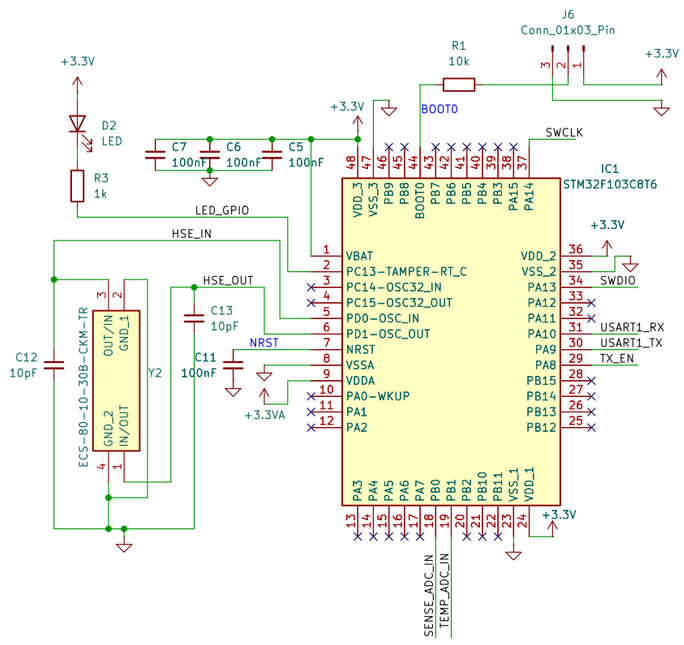
Secondary
3.3V Analog Supply



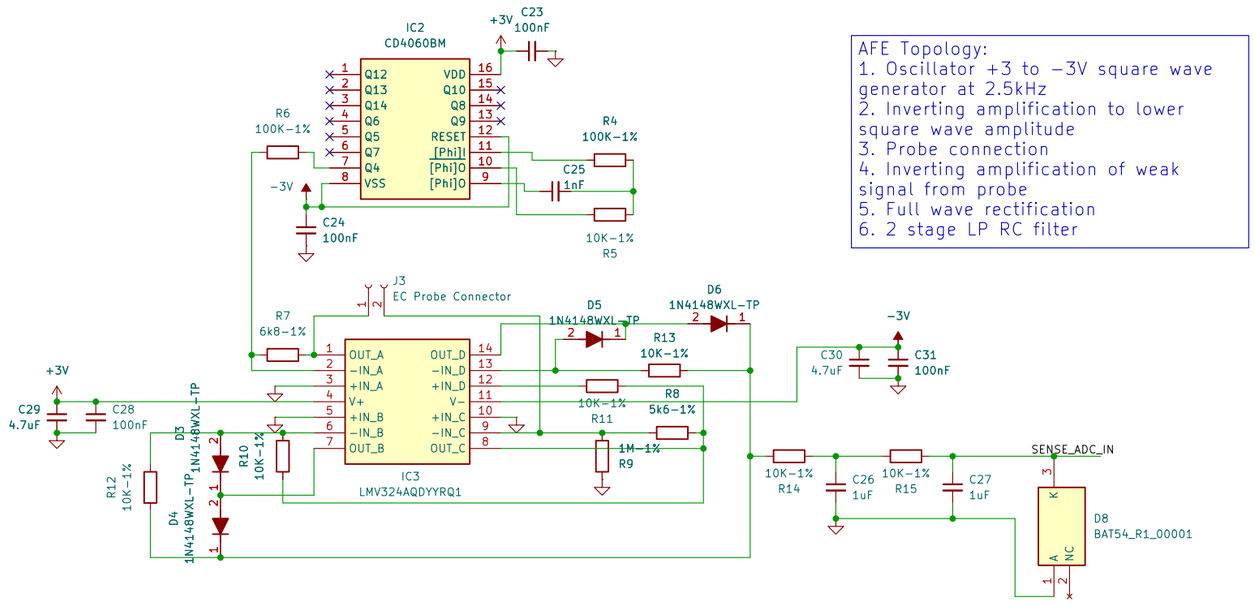
3V Regulator



−3V Charge Pump Inverter



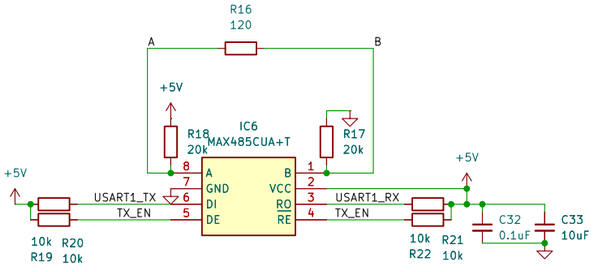
MCU(STM32)



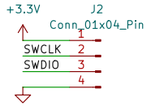
Electrical Conductivity Sensor Analog Front End

AFE Topology:
1. Oscillator +3 to −3V square wave generator at 2.5kHz
2. Inverting amplification to lower square wave amplitude
3. Probe connection
4. Inverting amplification of weak signal from probe
5. Full wave rectification
6. 2 stage LP RC filter

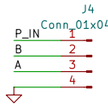
RS485 XCVR



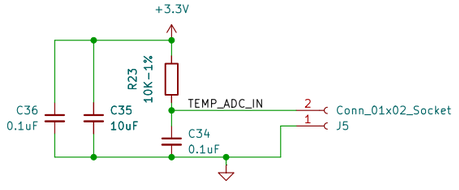
JTAG Connector



RS485 Connector



10k Thermistor Sensor Circuit



Rev	
A	1. Updated U1 Symbol to match datasheet pins. 2. Updated U2 Symbol to match datasheet pins. 3. Update R5 from 100k to 10k resistor

JEHL Solutions

Sheet: /
File: TDS_Sensor.kicad_sch

Title: Total Dissolved Solids Sensor

Size: A3	Date: 2026-11-01	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/1